

School of Computing and Artificial Intelligence
Faculty of Engineering and Technology
Sunway University

SEMESTER	April 2026
COURSE NAME	CSC3206 Artificial Intelligence
TITLE	Assignment 1
LECTURER	Dr. Teoh Yun Xin
PROGRAMME	Bachelor of Information Systems (Honours) (Data Analytics)
GROUP NAME	Giggle
GROUP MEMBER	1. Wong Hui San (22034540) 2. Ayu Wen Li (22017867) 3. Keertana A/P Subramaniam (23109614) 4. Siow Qi Yung (22053037) 5. Vidhipriya Devi Pokiri (23057953)
SUBMISSION DATE	21 st June 2026, 23:59

Table of Contents

1.0 Problem Definition and Data Collection	3
1.1 Problem Definition	3
1.2 Data Collection	4
2.0 Problem Analysis	10
2.1 State Space	10
2.2 Initial State	10
2.3 Goal State	10
2.4 Actions	11
2.5 Transition Model	12
2.6 Path Cost Function	12
3.0 AI Algorithm Evaluation	15
3.1 Breadth-First Search	15
3.2 Uniform Cost Search	18
3.3 A* Search	23
4.0 Algorithm Comparison and Conclusion	29
4.1 Algorithm Comparison	29
4.2 Conclusion	31
5.0 Appendices	32
5.1 Video Presentation	32
5.2 Individual Reflection	32
Wong Hui San (22034540)	32
Ayu Wen Li (22017867)	34
Keertana A/P Subramaniam (23109614)	36
Siow Qi Yung (22053037)	38
Vidhipriya Devi Pokiri (23057953)	40
5.3 Evidence of Group Collaboration	42
5.4 AI Usage Transparency	44
6.0 References	45

1.0 Problem Definition and Data Collection

1.1 Problem Definition

This assignment focuses on designing a house visit tour that involves route planning starting from Sunway University and visiting the residential areas of five members. As there is more than one possible route that can be generated, AI search algorithms are used to determine the best path. One of the criteria used to evaluate the best path is the carbon emission produced throughout the journey, which relates to United Nations SDG 13: Climate Action. According to the Malaysia Transport and Climate Profile 2026, Malaysia's transport emissions per GDP are higher than the regional average. Reducing transport emissions and improving fuel efficiency are part of Malaysia's efforts to achieve a low-carbon economy and national climate targets (Asian Transport Observatory). Therefore, the final route chosen must produce least carbon emissions.

Another constraint in this route planning is the financial constraint. As all members are currently university students, the budget is limited and the total travel expenses needs to be minimised. This includes reducing fuel expenses, toll charges, public transportation ticket fees, and other related expenses incurred during the trip.

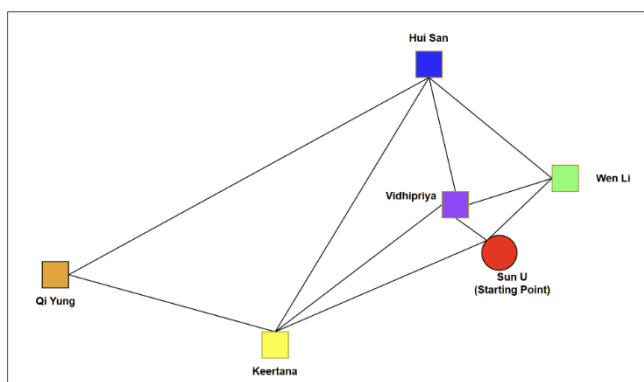
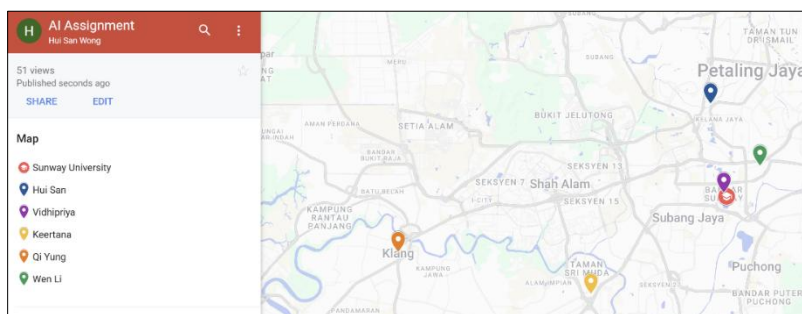
There are a few assumptions made in order to find the best path. All traffic conditions are assumed to be normal with no heavier than usual congestion and fuel prices are assumed to be subsidised and remain constant. Furthermore, all road connections between locations are assumed to be accurately represented using Google Maps, and the distances and travel times obtained are reliable. Lastly, the same type of car, a Perodua Axia, is assumed to be used for the driving mode.

1.2 Data Collection

a) Group Member Locations

Label	Starting Point	
Sun U	Sunway University	5, Jln Universiti, Bandar Sunway, 47500 Petaling Jaya, Selangor
Label	Member	Residential Area
Hui San	Wong Hui San	Taman Mayang, 47301 Petaling Jaya, Selangor
Wen Li	Ayu Wen Li	Petaling Jaya South 5, 46000 Petaling Jaya, Selangor
Keertana	Keertana A/P Subramaniam	Kemuning Utama, 40460 Shah Alam, Selangor
Qi Yung	Siow Qi Yung	Klang, 41300 Klang, Selangor
Vidhipriya	Vidhipriya Devi Pokiri	Bandar Sunway, 47500 Subang Jaya, Selangor

Google Maps is the main tool used for obtaining distance values and suggested routes. In this route planning context, a neighbour node means any node that can be reached directly from the current node without passing through any intermediate nodes. The base graph is constructed by plotting all members' residential areas and the starting point, Sunway University. Then, Draw.io is used to construct the base graph by replicating the locations and drawing edges to connect the nodes.



b) Carbon Emission Data

i) Car Travel Mode

The formula used for the car travel carbon emission is:

$$\text{Emission} = \text{Distance (km)} \times \text{Car Emission Factor (kg CO}_2\text{e/km)} \div \text{Passenger Count}$$

In terms of car transportation, assumptions were made for generalisation such as that the assumed fuel type is Petrol (RON95). Thus, based on the fuel type the assumed average fuel consumption is 7ℓ/100km while petrol emission factor is 2.31 kg CO₂ per litre. Given that the formula for the car emission factor is average fuel consumption times petrol emission factor, the car emission factor will be 0.16 kg CO₂e/km. Additionally, to calculate the carbon emission of each person, the assumed number of passengers in the car is 1.

ii) Public Travel Mode

The formula used for the car travel carbon emission is:

$$\text{Emission} = \text{Distance (km)} \times \text{Average Vehicle Emission Factor (kg CO}_2\text{e/km)} \\ \div \text{Average Passenger Count}$$

Emission factor can be defined as the standard coefficient value used to quantify the rate of greenhouse gas pollutants released from a specific activity. The following emission rates for each of the public transport types has been calculated for each passenger per kilometre which results in the assumed average passenger occupancy rate to be 1. The emission factor for the average urban bus which uses diesel fuel is 0.08 kg CO₂e/km while an electric bus such as the Sunway BRT services has an emission factor of 0.02 kg CO₂e/km. Various type of commuter trains such as the LRT, MRT and KTM have an assumed emission factor of 0.04 kg CO₂e/km.

The following public transportation routes for each of the travel routes were planned to use the public transit mobile application called Moovit which helps to help the most efficient route to get to a specific destination. Each route can be a combination of LRT, standard urban bus, Sunway BRT or KTM.

From	To	Route	Formula (kg CO ₂ e)
Sun U	Vidhipriya	N/A (Walking)	0 kg CO₂e
Sun U	Wen Li	Bus Desa Mentari Fasa 4 (Utara) to Bus Kompleks 3c (8.4km)	8.4 km × 0.08 kg CO ₂ e/km = 0.672 kg CO₂e
Sun U	Keertana	LRT Alam Megah to LRT USJ 7 (5.4km) BRT USJ7 to BRT SunU-Monash (2.2km)	5.4km (0.04 kg CO ₂ e/km) + 2.2km (0.02 kg CO ₂ e/km) = 0.26 kg CO₂e
Vidhipriya	Hui San	LRT Lembah Subang to LRT Kelana Jaya (1.4km) Bus LRT Kelana Jaya to Bus Kompleks 3c (11.8km)	1.4km (0.04 kg CO ₂ e/km) + 11.8km (0.08 kg CO ₂ e/km) = 1 kg CO₂e
Vidhipriya	Wen Li	Bus Sunway Clio (Opp) to Bus Menara Ann Joo (Opp) (5km)	5km × 0.08 kg CO ₂ e/km = 0.4 kg CO₂e
Vidhipriya	Keertana	BRT Sun U-Monash to BRT Setia Jaya (3.2km) KTM Setia Jaya to Stesen KTMB Shah Alam (11km)	3.2km (0.02 kg CO ₂ e/km) + 11km (0.04 kg CO ₂ e/km) = 0.504 kg CO₂e
Wen Li	Hui San	Bus Desa Mentari Fasa 4 (Utara) to Bus KJMC (8.5km)	8.5km (0.08 kg CO ₂ e/km) = 0.68 kg CO₂e
Keertana	Hui San	LRT Alam Megah to LRT Taman Bahagia (17.9km) Bus Taman Bahagia to Bus SMK Kelana Jaya (2km)	17.9km (0.04 kg CO ₂ e/km) + 2km (0.08 kg CO ₂ e/km) = 0.876 kg CO₂e
Keertana	Qi Yung	Bus SA408 Komersial Seksyen 25 to Bus Hentian Klang (BSN) (15.5km)	15.5km (0.04 kg CO ₂ e/km) = 0.62 kg CO₂e
Hui San	Qi Yung	LRT Lembah Subang to LRT Subang Jaya (6.5km) KTM Subang Jaya to KTM Teluk Pulai (16.5km)	6.5km (0.04 kg CO ₂ e/km) + 16.5km (0.04 kg CO ₂ e/km) = 0.92 kg CO₂e

Carbon Emission					
From	To	Distance	By Car (per passenger)	By Public Transport (per passenger)	Walking
Sun U	Vidhipriya	0.4 km	Walking	Walking	0 emissions
Sun U	Wen Li	3.9 km	0.624 kg CO_2e	0.672 kg CO_2e	
Sun U	Keertana	13 km	2.080 kg CO_2e	0.26 kg CO_2e	
Vidhipriya	Hui San	8.8 km	1.408 kg CO_2e	1 kg CO_2e	
Vidhipriya	Wen Li	5.3 km	0.848 kg CO_2e	0.4 kg CO_2e	
Vidhipriya	Keertana	13 km	2.080 kg CO_2e	0.504 kg CO_2e	
Wen Li	Hui San	8.2 km	1.312 kg CO_2e	0.68 kg CO_2e	
Keertana	Hui San	19 km	3.040 kg CO_2e	0.876 kg CO_2e	
Keertana	Qi Yung	18 km	2.880 kg CO_2e	0.62 kg CO_2e	
Hui San	Qi Yung	24 km	3.840 kg CO_2e	0.92 kg CO_2e	

c) Travel Expenses Data

There are a few assumptions made when calculating the cost for different transportation modes. For driving a private car, the fuel price is assumed to be RM1.99 per litre. A Perodua Axia is used as a reference for the calculations. The fuel efficiency of the Perodua Axia is assumed to be 7L per 100km (Jax, 2025). Therefore, the formula below is used to calculate fuel consumption:

$$\text{Fuel Consumption (Litre)} = \text{Distance (km)} \times 0.07$$

To calculate fuel costs, fuel consumption is multiplied by the fuel price:

$$\text{Fuel Expense (RM)} = \text{Fuel Consumption (Litre)} \times \text{Fuel Price (RM/Litre)}$$

The total cost of travelling by car is the sum of the fuel cost and toll charges.

$$\text{Total Travel Expenses (RM)} = \text{Fuel Cost (RM)} + \text{Toll Charges (RM)}$$

As for public transport, the expense refers to the price that needs to be borne by passengers when using public transport. Therefore, the fares for different public transport modes are taken from online resources such as MyRapid and are assumed to be accurate.

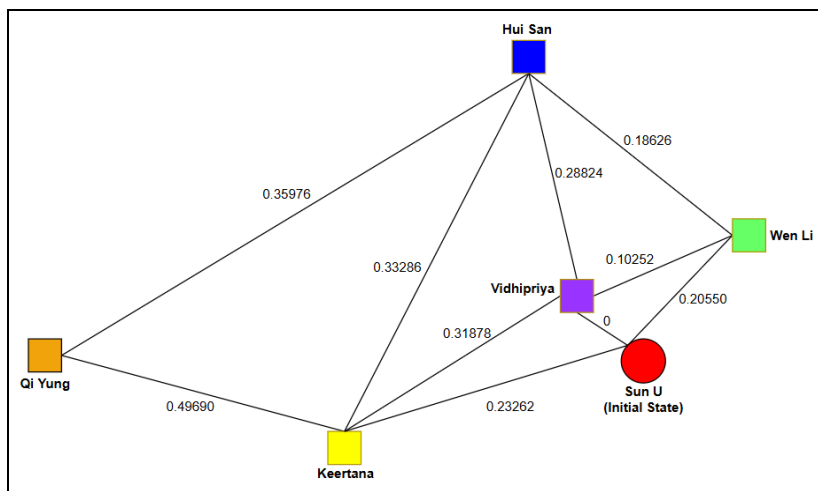
		By Car			By Public Transport	By Walking
From	To	Distance (km)	Fuel consumption (L)	Total Travel Expenses (RM)	Fare (RM)	Fare (RM)
Sun U	Vidhipriya	0.4	0.028	Walking	Walking	Total: 0
Sun U	Wen Li	3.9	0.273	Fuel cost: 0.54 Toll: 2.10 Total: 2.64	1.50 Total: 1.50	
Sun U	Keertana	3.9	0.910	Fuel cost: 1.81 Toll: 2.50 Total: 4.31	2.00(LRT) 2.70(BRT) Total:4.70	

Vidhipriya	Hui San	8.8	0.616	Fuel cost: 1.23 Toll: 3.10 Total: 4.33	1.20(LRT) 2.00 (Bus) Total:3.20	
Vidhipriya	Wen Li	5.3	0.371	Fuel cost: 0.73 Total: 0.73	1.00 Total:1.00	
Vidhipriya	Keertana	13.0	0.910	Fuel cost: 1.81 Toll: 2.50 Total: 4.31	3.60(BRT) 2.30(KTM) Total:5.90	
Wen Li	Hui San	8.2	0.574	Fuel cost: 1.13 Total: 1.13	2.00 Total: 2.00	
Keertana	Hui San	19.0	1.330	Fuel cost: 2.64 Total: 2.64	3.80(LRT) 1.00(Bus) Total: 4.80	
Keertana	Qi Yung	18.0	1.260	Fuel cost: 2.51 Toll: 1.56 Total: 4.07	9.80(Bus) Total: 9.80	
Hui San	Qi Yung	24.0	1.680	Fuel cost: 3.34 Total: 3.34	1.80(LRT) 3.50(KTM) Total: 5.30	

2.0 Problem Analysis

2.1 State Space

The state space is represented as a weighted graph where nodes represent group members' residences and edges represent possible routes between residences based on Google Maps connections. Each edge contains a path cost value calculated using factors including carbon emission and travel expenses. The path cost function is focusing on carbon emission and travel expenses, which is $Path\ Cost = 0.6(Carbon\ Emission) + 0.4(Travel\ Expenses)$. Diagram below shows the graph representation of the state space.



2.2 Initial State

The initial state of the problem can be defined as the starting location which in this assignment context is Sunway University. At this stage, the residential locations of each group member have not been visited yet and the route has not been determined. Besides that, no transportation actions have been taken, while the total path cost which includes the travel expenses, fuel consumption, and carbon emissions are all zero.

2.3 Goal State

The goal state is defined as the state where the route visited all the houses at least once starting from the root node which is Sunway University. Residential locations of Vidhipriya, Wen Li, Hui San, Keertana and Qi Yung must be visited at least once so that it can be considered as a complete route which is generated by the AI search algorithm. The best algorithm should generate a route where the overall path cost should be minimal so that it can satisfy the sustainability and the budget considerations.

2.4 Actions

Actions are the fundamental movements or transitions between geographic locations (nodes) that the AI search algorithm uses to build a complete path from the Initial State to the Goal State. An action represents a transition from a Current Location (N_i) to a Neighboring Location (N_j) using a designated Transport Mode (T).

- Locations (Nodes): {Sun U, Hui San, Wen Li, Keertana, Qi Yung, Vidhipriya}
- Transport Modes: Driving (Car), Public Transport (LRT, BRT, Bus, KTM), or Walking

The successful execution of an action updates the state using the Transition Model, adding the new location (N_j) to the set of visited locations.

The action space is the union of all feasible movement-mode combinations between connected node pairs. The AI search algorithm must evaluate all available transport modes for each link to identify the action with the lowest Path Cost. The connections are categorized as follows:

1. Walking Only: The link between Sun U $\leftrightarrow$ Vidhipriya only allows walking, which results in zero carbon emission and zero travel expenses.
2. Driving or Public Transport: The remaining 9 links are bidirectional and offer a choice between Driving (Car) and Public Transport, leading to four possible actions for each link (two directions, two modes).

The 9 connected links are:

- Sun U $\leftrightarrow$ Wen Li
- Sun U $\leftrightarrow$ Keertana
- Vidhipriya $\leftrightarrow$ Hui San
- Vidhipriya $\leftrightarrow$ Wen Li
- Vidhipriya $\leftrightarrow$ Keertana
- Wen Li $\leftrightarrow$ Hui San
- Keertana $\leftrightarrow$ Qi Yung
- Keertana $\leftrightarrow$ Hui San
- Hui San $\leftrightarrow$ Qi Yung

The AI search algorithm's primary function is to select the optimal sequence of these actions to satisfy the Goal State while minimizing the accumulated Path Cost.

2.5 Transition Model

The transition model can be defined as the function that outlines the result of moving from one state to another and an action into a new state which is $f(\text{current state, action}) = \text{next state}$. The transition model will show how the agent moves from one residence (state) to another after an action is performed. Each state is represented as (current location, visited set). The model will also update the current residence (state) and will mark the residence as already visited. The path cost is updated based on transportation cost and carbon emissions.

General transition model:

$$(\text{current location, visited set}) \rightarrow (\text{new location, visited set} \cup \{\text{new location}\})$$

Example based on problem scenario:

$$(\text{Wen Li, } \{\text{Vidhipriya, Wen Li}\}) \rightarrow (\text{Hui San, } \{\text{Vidhipriya, Wen Li, Hui San}\}).$$

Current State = Wen Li

Visited Residences = Vidhipriya and Wen Li

Action = move to Hui San by any transportation mode either by car or public transport

New Location = Hui San

This transition shows that the agent is currently at Wen Li and has already visited Vidhipriya and Wen Li. After performing the action of moving to Hui San using any available transportation mode, the agent's location is updated to Hui San and Hui San is added to the visited set.

2.6 Path Cost Function

The path cost function is used to determine the quality of each possible route. In this case, two factors are considered, which are carbon emission produced and travel expenses. These two factors are used to construct the path cost function and combined into a single comparable value. The final path cost is calculated as a weighted sum, where carbon emission is given a higher weight as this is the main objective of the path, reflecting environmental impact, while cost is also considered for affordability:

$$\text{Path Cost} = 0.6(\text{Carbon Emission}) + 0.4(\text{Travel Expenses})$$

However, carbon emission and travel expenses are in different units. Therefore, it is necessary to first normalise the values of both factors to make sure that they are on the same scale and do not dominate the calculation.

$$X_{\{normalized\}} = \frac{(X - X_{\{minimum\}})}{(X_{\{maximum\}} - X_{\{minimum\}})}$$

Normalisation for Carbon Emission:

0 was chosen as the X_{minimum} because it is the lowest carbon emission travel from Sun U to Vidhipriya by walking. Whereas, 3.840 was chosen as the X_{maximum} because the carbon emission produced when travelling from Hui San to Qi Yung by public transport is the highest.

$$X_{\{normalized\}} = \frac{(X - 0)}{(3.840 - 0)}$$

From	To	Carbon Emission (kg CO2e)	Normalised Carbon Emission (kg CO2e)
Sun U	Vidhipriya	0 (By Walking)	0
Sun U	Wen Li	0.624 (By Car)	0.1625
Sun U	Keertana	0.26 (By Public Transport)	0.0677
Vidhipriya	Hui San	1 (By Public Transport)	0.2604
Vidhipriya	Wen Li	0.4 (By Public Transport)	0.1042
Vidhipriya	Keertana	0.504 (By Public Transport)	0.1313
Wen Li	Hui San	0.68 (By Public Transport)	0.1771
Keertana	Hui San	0.876 (By Public Transport)	0.2281
Keertana	Qi Yung	0.62 (By Public Transport)	0.1615
Hui San	Qi Yung	0.92 (By Public Transport)	0.2396

Normalisation for Travel Expenses:

The X_{minimum} represents the lowest cost to travel from one point to another. In this case, RM0 is the lowest cost, which occurs when travelling from Sun U to Vidhipriya by walking. Whereas, the X_{maximum} represents the highest cost, which is travelling from Keertana to Qi Yung by public transport. In this case, it is RM9.80 for the bus fare.

$$X_{\{normalized\}} = \frac{(X - 0)}{(9.80 - 0)}$$

From	To	Travel Expenses (RM)	Normalised Travel Expenses (RM)
Sun U	Vidhipriya	0 (By walking)	0
Sun U	Wen Li	2.64 (By car)	0.27
Sun U	Keertana	4.70 (By public transport)	0.48
Vidhipriya	Hui San	3.20 (By public transport)	0.33
Vidhipriya	Wen Li	1.00 (By public transport)	0.10
Vidhipriya	Keertana	5.90 (By public transport)	0.60
Wen Li	Hui San	2.00 (By public transport)	0.20
Keertana	Hui San	4.80 (By public transport)	0.49
Keertana	Qi Yung	9.80 (By public transport)	1.00
Hui San	Qi Yung	5.30 (By public transport)	0.54

Calculation for Path Cost:

The normalised values are substituted into the formula to calculate the path cost.

$$Path\ Cost = 0.6(Carbon\ Emission) + 0.4(Travel\ Expenses)$$

From	To	Path Cost
Sun U	Vidhipriya	0
Sun U	Wen Li	0.20550
Sun U	Keertana	0.23262
Vidhipriya	Hui San	0.28824
Vidhipriya	Wen Li	0.10252
Vidhipriya	Keertana	0.31878
Wen Li	Hui San	0.18626
Keertana	Hui San	0.33286
Keertana	Qi Yung	0.49690
Hui San	Qi Yung	0.35976

3.0 AI Algorithm Evaluation

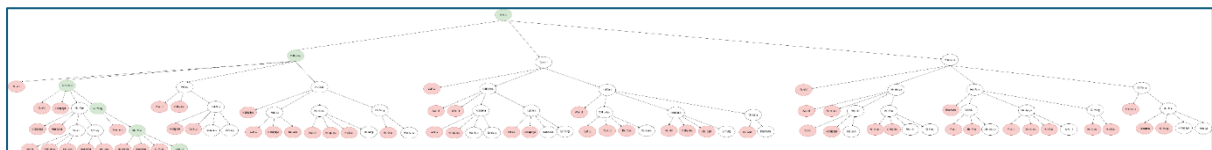
3.1 Breadth-First Search

Breadth-First Search (BFS) is an uninformed search algorithm that explores the search tree level by level. The algorithm uses a First-In First-Out (FIFO) queue to store nodes that need to be explored. BFS visits all neighbouring nodes of the current state first before moving to deeper levels. This approach guarantees that the first solution found contains the minimum number of steps but does not consider the path cost.

Implementation of Breadth-First Search:

In this route planning problem, the initial state is Sun U and the goal state is to visit all five residences at least once. BFS begins by expanding all residences directly connected to Sun U, which is Vidhipriya, Wen Li and Keertana. These residences form the first level of the search tree. The algorithm then continues to expand nodes according to the order in which they were added to the frontier queue. Starting from Vidhipriya, BFS explores its connected residences before moving to Wen Li and Keertana. This process continues until a state is reached where all five residences have been visited. Lastly, the first possible solution path obtained using BFS is Sun U $\rightarrow$ Vidhipriya $\rightarrow$ Keertana $\rightarrow$ Qi Yung $\rightarrow$ Hui San $\rightarrow$ Wen Li, with a total path cost of 1.3617 km.

In the Breadth-First Search tree diagram, the nodes in green colour show the goal states where all nodes have been visited, while the nodes in red colour show loops which is the redundant path that were removed during the search process.



For a clearer view of the diagram, please click the link below and open the diagram with draw.io:

[https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Graph%2FMap%20\(AI%20Assignment\).drawio&page-id=dOcEEuVGubM_NfwaXgWc&dark=0#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1z3ukS7GKiOe1GUBnFeuh6HYC3BFxustA%26export%3Ddownload](https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Graph%2FMap%20(AI%20Assignment).drawio&page-id=dOcEEuVGubM_NfwaXgWc&dark=0#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1z3ukS7GKiOe1GUBnFeuh6HYC3BFxustA%26export%3Ddownload)

Evaluation of Breadth-First Search:

a) Completeness and Optimality

Breadth-First Search is a complete search algorithm because it explores all possible paths level by level, so if a solution exists, it will always find a solution. Since the number of residences in the network is finite, BFS will explore all possible routes and reach the goal state. However, BFS is only optimal when every path has the same cost. In this route planning problem, the path cost between residences is different. Therefore, BFS can find a route that uses the fewest number of visits, but it does not guarantee the lowest carbon emission and travel expenses.

b) Time and Space Complexity

Breadth-First Search does not require too much time or memory since the number of residences is small. Its time complexity is $O(b^d)$, where b is the number of choices such as how many nearby residences each node has and d is how many steps are needed to reach the solution. This means that if the network has more residences, BFS will need to check many more possibilities. BFS does not use much memory because the search space is small. However, when the number of residences increases, the amount of memory needed also increases, which will make BFS less efficient for large route planning problems. In our case, the network is small and BFS can still work efficiently without causing major performance issues.

c) Practical Feasibility

Breadth-First Search is quite easy to implement and easy to understand because it follows a simple rule which is it explores all nearby nodes first before moving further. It is useful at finding a solution quickly in terms of the number of steps, so it can work well for small route planning problems where the network is not too big. However, BFS becomes less practical when the network is large. It needs to store many possible paths in memory while searching. Also, it does not consider the actual path cost such as travel distance, cost and carbon emission when choosing which path to explore. As a result, it may find a route that uses fewer steps but the path cost of the route may be high.

d) Sustainability Considerations

Breadth-First Search does not consider any path cost when selecting routes including travel distance, travel expenses, carbon emission and fuel consumption. Therefore, it may generate routes that require more fuel and lead to higher transportation costs. Longer travel distances can also result in increased carbon emissions, which negatively affect environmental

sustainability. Consequently, BFS does not support the goals that are related to environmental sustainability, such as the United Nations Sustainable Development Goals (SDG) 13: Climate Action.

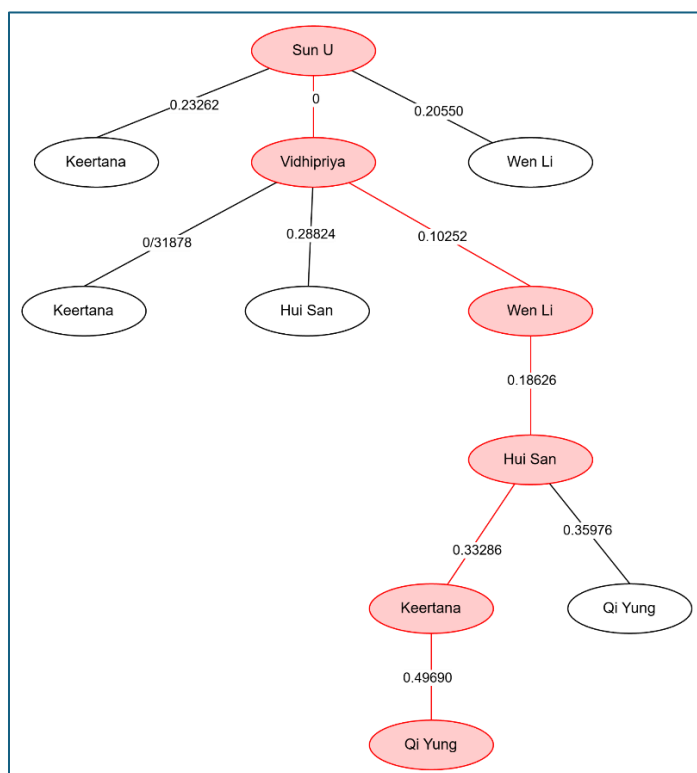
Overall, Breadth-First Search is useful when we want minimum number of visits and to provide a simple baseline solution. However, it is less suitable when the goal is to minimize distance, fuel consumption, travel expenses and carbon emissions or when the network is large.

3.2 Uniform Cost Search

Uniform Cost Search is an uninformed artificial intelligence search algorithm that focuses on finding the lowest cost path from the start to end node in a graph. This algorithm works by using the concept of priority queue in order to expand and store the weighted graph nodes with the least cost (Geeksforgeeks, 2025).

During the initial stages at the root node, the cumulative path cost will be zero. Once the nodes with the lower path cost are expanded, the cumulative cost of the expanded nodes will be calculated, prioritising and removing the expanded nodes from the priority queue while the neighbour nodes are expanded. The sum of path cost from the start to current node of each neighbour node will be totalled. The neighbour node that is not in the priority queue will be added in with the calculated cost while the algorithm compares between the updated and previously recorded cost of the neighbour nodes in the queue. In order to ensure that the route is cost effective, the cost will be updated and the path recorded once a cheaper path is found. The process is repeated until the goal node has been reached and the search process is terminated. Using the cumulative cost of reaching the goal node, the algorithm will reconstruct the path by retracing the nodes.

Implementation of Uniform Cost Search:



Start at Sun U:

Start Node: {Sun U}

Unvisited Nodes: {Vidhipriya, Wen Li, Hui San, Keertana, Qi Yung}

Path Cost	Path Leads to the Leaf Node
0	Sun U

After Expanding Sun U:

From	To	Edge Path Cost	Cumulative Path Cost
Sun U	Vidhipriya	0	0
Sun U	Keertana	0.23262	0.23262
Sun U	Wen Li	0.20550	0.20550

Lowest cumulative cost: $\min(0, 0.20550, 0.23262) = 0$

Selected Node: Vidhipriya

Current Route: Sun U $\rightarrow$ Vidhipriya

After Expanding Vidhipriya:

Unvisited Nodes: {Wen Li, Hui San, Keertana, Qi Yung}

Path Cost	Path Leads to the Leaf Node
0	Sun U - Vidhipriya

From	To	Edge Path Cost	Cumulative Path Cost
Vidhipriya	Keertana	0.31878	0.31878
Vidhipriya	Wen Li	0.10252	0.10252
Vidhipriya	Hui San	0.28824	0.28824

Lowest cumulative cost: $\min(0.10252, 0.28824, 0.31878) = 0.10252$

Selected Node: Wen Li

Current Route: Sun U $\rightarrow$ Vidhipriya $\rightarrow$ Wen Li

After Expanding Wen Li:

Unvisited Nodes: {Hui San, Keertana, Qi Yung}

Path Cost	Path Leads to the Leaf Node
-----------	-----------------------------

0.10252	Sun U - Vidhipriya - Wen Li
---------	-----------------------------

From	To	Edge Path Cost	Cumulative Path Cost
Wen Li	Hui San	0.18626	0.28878

Selected Node: Hui San

Current Route: Sun U → Vidhipriya → Wen Li → Hui San

After Expanding Hui San:

Unvisited Nodes: {Keertana, Qi Yung}

Path Cost	Path Leads to the Leaf Node
0.28878	Sun U - Vidhipriya - Wen Li - Hui San

From	To	Edge Path Cost	Cumulative Path Cost
Hui San	Keertana	0.33286	0.62164
Hui San	Qi Yung	0.35976	0.64854

Lowest cumulative cost: $\min(0.62164, 0.64854) = 0.62164$

Selected Node: Keertana

Current Route: Sun U → Vidhipriya → Wen Li → Hui San → Keertana

After Expanding Keertana:

Unvisited Nodes: {Qi Yung}

Path Cost	Path Leads to the Leaf Node
0.28878	Sun U - Vidhipriya - Wen Li - Hui San - Keertana

From	To	Edge Path Cost	Cumulative Path Cost
Keertana	Qi Yung	0.49690	1.11854

Selected Node: Qi Yung

Final Optimal Route:

Final Route: Sun U → Vidhipriya → Wen Li → Hui San → Keertana → Qi Yung

Total path cost: $0 + 0.10252 + 0.18626 + 0.33286 + 0.49690 = 1.11854$

Evaluation of Uniform Cost Search:

a) Completeness

The UCS algorithm is complete as the path to the goal node will be found given that the solution exists and there is sufficient memory and time and that the edge costs are not negative. The implementation of uniform cost search in the house visit scenario is suitable as the path costs are guaranteed to be positive values as the values are calculated from the weighted carbon emissions and travel expenses, ensuring no residence will be accidentally left out.

b) Optimality

One of the advantages of uniform cost search is the optimality as the algorithm will ensure that the least costly path will be found to reach the goal state. Given that the path cost in this assignment is derived from carbon emission and travel expenses, the algorithm supports the goal of encouraging sustainability through reduced environmental impact as well as reducing transportation costs.

c) Time Complexity

Given that the uniform cost search algorithm explores the nodes with the least cost path, the time needed to expand the nodes may be inefficient especially if the state space is large. The formula for time complexity of UCS is $O(b^{1+\lceil \frac{C^*}{\epsilon} \rceil})$ where b is the branching factor while C^* refers to cost of the optimal solution and ϵ is the minimum edge cost. However, the time complexity is manageable in this assignment as there are only 6 locations to be explored. As the number of locations to be explored increases, so does the time complexity due to the growing requirement for memory resources

d) Space Complexity

Similar to the time complexity of UCS, space complexity may be an issue for this AI search algorithm in scenarios that concern many locations or state spaces. This is because every expanded node will be included into the priority queue, causing excessive memory load. However, in this assignment the number of residences to be explored is relatively small so excessive memory resources are not required.

e) Practical Feasibility

As the number of locations to be visited is relatively small, uniform cost search is a practical AI algorithm to implement in the house visit scenario especially with the objective of reducing carbon emission and travel expenses. The path cost is incorporated into the algorithm to select the less costly path. However, if the number of state spaces is large, heuristic-based algorithms would be more feasible compared to UCS as these algorithms take heuristic information into consideration in order to approximate the future costs and will in turn explore less nodes.

3.3 A* Search

A* Search is one of the informed AI search algorithms. One key difference between informed and uninformed search is the additional information about the goal state. Instead of blindly exploring nodes until the goal is reached, A* Search uses a heuristic function to estimate the cost from the current state to the goal state. This heuristic provides guidance during the search by prioritising nodes with the lowest estimated total cost. The evaluation function is represented as:

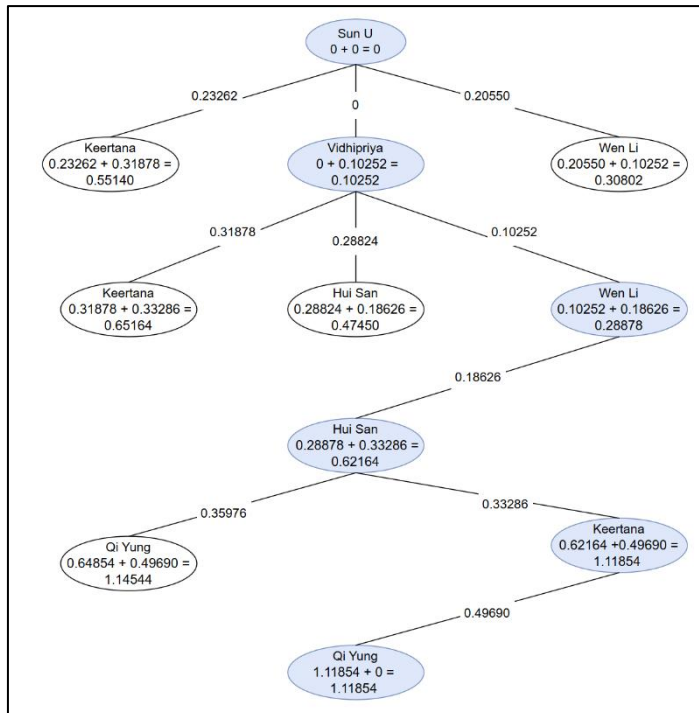
$$f(n) = g(n) + h(n)$$

In the context of this route planning problem, the variables are defined as follows:

- $f(n)$ is the estimated total cost of the cheapest solution path through node n
- $g(n)$ is the accumulated actual cost from the start node to the current node
- $h(n)$ is the heuristic estimate of the remaining cost to complete the route, it is estimated using the minimum path cost from the current node to any unvisited node

A* Search manages nodes between the Open list and Closed List. The Open List contains nodes that have been discovered but not yet expanded. At each step, the algorithm selects the node in the Open List with the lowest $f(n)$ for expansion, ensuring that the search is guided toward promising routes while still taking the total path cost into account. Meanwhile, Closed List contains nodes that have been expanded and evaluated. The node in the Closed List is not revisited unless a new route to that node has a lower $g(n)$ than the previously recorded route. This ensures that even though multiple routes exist, A* gradually constructs the most cost-effective path.

Implementation of A* Search:



Start at Sun U:

Open List = {Sun U}

Closed List = { }

- $g(\text{SunU}) = 0$
- $h(\text{SunU}) = \min\{\text{cost to each reachable neighbour}\} = \min\{0.23262, 0, 0.20550\} = 0$
- $f(\text{SunU}) = 0 + 0 = 0$

At the start, Sunway University is the only node in the Open List. Since no movement has occurred, $g(\text{SunU}) = 0$, and the heuristic $h(\text{SunU})$ is taken as the minimum cost to its neighbouring nodes, which is also 0. Therefore, $f(\text{SunU})=0$ and it is selected for expansion.

After Expanding Sun U:

Open List = {Vidhipriya, Wen Li, Keertana}

Closed List = {Sun U}

- To Keertana:
 - $g(\text{Keertana}) = 0.23262$
 - $h(\text{Keertana}) = \min\{0.31878, 0.33286, 0.49690\} = 0.31878$

- $f(\text{Keertana}) = 0.23262 + 0.31878 = 0.55140$
- To Vidhipriya:
 - $g(\text{Vidhipriya}) = 0$
 - $h(\text{Vidhipriya}) = \min\{0.10252, 0.28824, 0.31878\} = 0.10252$
 - $f(\text{Vidhipriya}) = 0 + 0.10252 = 0.10252$
- To Wen Li:
 - $g(\text{Wen Li}) = 0.20550$
 - $h(\text{WenLi}) = \min\{0.10252, 0.18626\} = 0.10252$
 - $f(\text{WenLi}) = 0.20550 + 0.10252 = 0.30802$

After expanding Sunway University, the node is moved from the Open List to the Closed List. Vidhipriya, Wen Li, and Keertana are placed in the Open List because they are neighbouring nodes of Sunway University and are available for future expansion. The algorithm evaluates all neighbouring nodes using the $f(n)$ value. Among them, Vidhipriya has the lowest $f(n)$, thus, it is selected as the next node to be expanded. The same procedure is repeated for all subsequent expansions until all residential areas are visited. The following calculations are summarized by just showing the $f(n)$ value instead of the full workings.

After Expanding Vidhipriya:

Open List = {Wen Li, Hui San, Keertana}

Closed List = {Sun U, Vidhipriya}

- To Keertana:
 - $f(\text{Keertana}) = 0.31878 + 0.33286 = 0.65164$
- To Hui San:
 - $f(\text{Hui San}) = 0.28824 + 0.18626 = 0.47450$
- To Wen Li:
 - $f(\text{Wen Li}) = 0.10252 + 0.18626 = 0.28878$

Wen Li has the lowest $f(n)$ value and is selected as the next node for expansion.

After Expanding Wen Li:

Open List = {Hui San}

Closed List = {Sun U, Vidhipriya, Wen Li}

- To Hui San:
 - $f(\text{Hui San}) = 0.28878 + 0.33286 = 0.62164$

Although Vidhipriya and Hui San are neighbours of Wen Li, only Hui San remains in the Open List because Vidhipriya has already been visited. Nodes in the Closed List are excluded to prevent revisiting. Therefore, Hui San is selected for expansion.

After Expanding Hui San:

Open List = { Keertana, Qi Yung }

Closed List = { Sun U, Vidhipriya, Wen Li, Hui San }

- To Qi Yung:
 - $f(\text{Qi Yung}) = 0.64854 + 0.49690 = 1.14544$
- To Keertana:
 - $f(\text{Keertana}) = 0.62164 + 0.49690 = 1.11854$

Keertana has the lower $f(n)$ and is selected for the next expansion.

After Expanding Keertana:

Open List = { Qi Yung }

Closed List = { Sun U, Vidhipriya, Wen Li, Hui San, Keertana }

- To Qi Yung:
 - $f(\text{Qi Yung}) = 1.11854 + 0 = 1.11854$

Since Qi Yung is the final destination, its heuristic value is 0. Qi Yung is selected, completing the search and forming the final optimal route.

Final Optimal Route:

Sun U → Vidhipriya → Wen Li → Hui San → Keertana → Qi Yung

Total Path Cost = $0 + 0.10252 + 0.18626 + 0.33286 + 0.49690 = 1.11854$

This is the route generated using A* Search. It results in the lowest total accumulated cost while making sure all locations are visited. The total path cost is calculated by adding up the individual path costs at each step along the route.

Evaluation of A* Search:

a) Completeness

A* Search is complete because it is guaranteed to find a solution whenever a valid solution exists (GeeksforGeeks, 2025). In the context of this assignment, A* Search will continue expanding and evaluating different location sequences until a valid path that connects all required locations is found. The search process prioritises more promising routes using the evaluation function, but it still ensures that all necessary possibilities are considered during exploration. As a result, the algorithm does not miss any potential valid solution and will eventually find a complete visiting route if one exists in the transportation network.

b) Optimality

A* search is optimal when the heuristic is admissible, meaning it never overestimates the actual cost (GeeksforGeeks, 2025). It always selects the lowest path cost based on both actual cost and estimated cost. The evaluation function ensures that decisions are not only based on the actual cost already travelled, but also consider the estimate of how much cost is still needed to travel from the current node to the goal node. In the context of this assignment, the heuristic uses the minimum path cost to an unvisited node, which is admissible because it will never overestimate the actual cost. Because of this admissible heuristic, A* ensures that the final route produced is the one with the lowest total combined cost, making it suitable for finding the most sustainable and affordable route.

c) Time Complexity

In the context of this assignment, the time complexity of A* Search depends on the number of locations (nodes) and possible routes (edges) in the network. In the worst case scenario, A* Search may need to explore an exponential number of possible paths. Each location can connect to multiple other locations. That is why the algorithm may need to explore several possible travel sequences before reaching the goal state where all houses are visited. However, a good heuristic based on carbon emissions and travel expenses can reduce the node exploration and significantly help in guiding the search toward the goal.

d) Space Complexity

A* Search has high space complexity in theory, which is $O(b^d)$, because it must store all generated but not yet expanded nodes in the Open List, and all expanded nodes in the Closed List (GeeksforGeeks, 2025). In the context of this assignment, each node represents a location,

and each move between locations creates a new possible route. As a result, multiple partial routes are stored in memory at the same time, making it memory intensive. However, in this assignment, there is only a small number of nodes, so the number of stored nodes is still manageable and practical for implementation. Therefore, it has medium space complexity in practice if the number of nodes remains small.

e) Practical Feasibility

From a practical feasibility perspective, A* Search is highly suitable for real-world route optimization problems because it can incorporate multi-factor cost functions like carbon emissions and financial expenses. It also efficiently guides the search toward better routes using heuristic estimation, rather than blindly exploring all possibilities. Since the number of nodes is small and the path costs are clearly defined using a weighted combination of carbon emissions and travel expenses, A* is easy to implement and effective in producing an optimal route.

In terms of broader impact, this aligns well with sustainability objectives and the United Nations Sustainable Development Goals (SDGs), particularly SDG 13 (Climate Action), as reducing carbon emissions supports climate change mitigation. At the same time, reducing travel expenses improves overall route efficiency and practicality in real-world applications.

One key factor that affects A* Search performance is the heuristic design. A weak heuristic can degrade efficiency, while a strong heuristic can significantly improve performance (Kumari, 2025). In conclusion, A* Search provides a strong balance between optimality and efficiency in route planning problems.

4.0 Algorithm Comparison and Conclusion

4.1 Algorithm Comparison

Algorithms	Completeness	Optimality	Time Complexity	Space Complexity	Practical Feasibility
BFS	Yes. Guaranteed since the state space of 6 residences is finite.	No. Ignores the weighted path cost function and selects the shortest path by steps, not cost, resulting in a suboptimal path cost of 1.3617.	Time-consuming for large networks. Blind search, exploring cost-irrelevant paths.	Stores all generated nodes, leading to high memory consumption.	Only suitable for very small, unweighted networks where path cost is irrelevant. Not a viable choice for optimization problems.
UCS	Yes. Guaranteed since all edge Path Costs are positive (derived from weighted CE and expenses).	Yes. Guaranteed to find the least costly path and achieved the proven optimal Path Cost of 1.11854.	Performance can degrade rapidly if costs are large. Does not use heuristic guidance to prune paths.	High. As every expanded node is stored in the priority queue and is manageable for small, 6-node size.	Practical for cost-minimization goals in small-scale networks. Less scalable than A* search for complex road networks.
A* Search	Yes. Guaranteed to find a solution when one exists.	Yes. Guaranteed to be optimal due to the use of an admissible approximation. Achieved the lowest total combined Path Cost of 1.11854.	Efficient. Guided by heuristic function and typically explores fewer nodes than UCS/BFS in practice, improving speed.	Highly feasible, practical search space is reduced by the approximations and medium memory complexity for small problems.	Most suitable for multi-factor real-world optimization problems, offering a balance between efficiency and optimality.

i) Efficiency

A* Search has the most efficiency compared to BFS and UCS due to its heuristic function. This function is used to guide the search so it can find a route that is most likely to lead to a lower overall cost. BFS usually explores the nodes by levels and does not consider the path costs. At the same time, UCS only considers the cost that was accumulated from the start node. Whereas A* Search combines both the actual path cost and the estimated future cost by using the evaluation function: $f(n) = g(n) + h(n)$.

The evaluation function helps A* Search to prioritize routes that will most likely lead to optimal solutions and also reduce node expansions that are unnecessary. This can significantly reduce search time and the effort for computation when applying this algorithm in larger transportation networks unlike BFS and UCS. A* Search is the most scalable approach although the locations used are in small numbers.

ii) Accuracy of Solution

A* Search algorithm achieved the objective which is to minimize the overall path cost when it successfully produced the lowest-cost route as shown below:

Sun U → Vidhipriya → Wen Li → Hui San → Keertana → Qi Yung

Total Path Cost = $0 + 0.10252 + 0.18626 + 0.33286 + 0.49690 = 1.11854$

UCS achieved the same optimal route as it also focuses on minimizing the path cost. However, BFS obtained a higher total path cost of 1.36170 compared to A* Search and UCS. This shows that BFS does not optimize for sustainability or cost. Additionally, A* Search used heuristic which is based on the minimum path cost to an unvisited node. Hence, it does not overestimate the remaining cost. Therefore, A* Search is able to maintain the optimality and improve the search efficiency at the same time as the heuristic is admissible.

iii) Real-World Applicability

A* Search is an algorithm that is widely used in route planning and navigation systems in the real-world. This is due to the algorithm's ability to incorporate multiple optimization factors simultaneously. The path cost function used in this assignment considers both environmental

and economic factors. This helps to identify routes that are not only cheap but also environmentally sustainable. Furthermore, as A* Search uses heuristic guidance, the algorithm is more suitable to be used in larger route planning where it involves more locations, transportation modes, and constraints. A* Search is the most suitable algorithm that can be applied to real-world applications. This is because it helps with route planning while also considering travel expenses, distance, and environmental impact.

iv) Problem Constraints

The problem constraints in this assignment are as below:

- Start node is from Sunway University.
- All residential locations must be visited at least once.
- The combined path cost must be minimal.
- Carbon emissions and travel expenses are considered.
- Transportation network operations must be connected.

A* Search satisfies all the constraints mentioned above. It successfully visited all the residential locations while minimizing the total path cost produced. A* Search directly optimizes the function of the objective as defined in the formulation of the problem unlike BFS which only focuses on the number of steps. Although UCS has the same total path cost, A* Search is the more practical choice as it can benefit from the heuristic guidance when the size of the problem increases.

4.2 Conclusion

In summary, A* Search is the most suitable algorithm as it offers the best balance between computational efficiency, optimality and practical applicability although UCS achieved the same optimal route. The total path cost obtained by the algorithm which is 1.11854 satisfied all the constraints and at the same time implemented both carbon emissions and travel expenses into the process of decision-making effectively. Hence, A* Search is selected as the final best recommended algorithm when it comes to generating sustainable and expenses-effective residential visitation routes. This algorithm also contributes towards SDG 13 which focuses on Climate Action by reducing the carbon emissions by using public transportations.

5.0 Appendices

5.1 Video Presentation

The following link directs to our group video presentation, which is approximately 10 minutes long. The video is uploaded to Google Drive and can be viewed by anyone with the link.

<https://drive.google.com/file/d/1XKKTEaxHA70x3bpfm-sacDMRoosWGIkx/view?usp=sharing>

5.2 Individual Reflection

Wong Hui San (22034540)

In this assignment, I took on the role of team leader. I led the initial planning by reviewing requirements and preparing a task delegation plan with internal deadlines. I also organized group discussions and consulted the lecturer whenever needed. My main contributions included developing the problem definition by designing the scenario for the assignment. I was also responsible for collecting data on travel expenses across all routes. In the AI formulation section, I constructed the path cost function using a weighted approach. I also contributed to the evaluation of the A* Search algorithm by explaining its working process, demonstrating manual calculations, and discussing its suitability, and presented this section in the video presentation.

Throughout this assignment, I learned that AI search algorithms can be applied to solve real-world problems, which I found fascinating. Previously, I viewed these algorithms as mainly theoretical or computing-based, but this project helped me understand their practical relevance. While working on the A* Search algorithm calculations, I found out that the algorithm was not as complicated as I initially thought. As long as the required information and values are clearly defined, the calculations can be performed systematically.

One challenge we faced was constructing the graph using real-world locations. Initially, we were unsure how to define neighbouring nodes and establish correct connections between locations. We overcame this by having a physical meeting where everyone shared ideas and discussed possible graph structures, allowing us to agree on a suitable representation. Another challenge was data collection due to mixed transportation modes with different carbon emissions and travel expenses. To manage this, we first identify the transportation mode for each route. Then, we calculated carbon emissions, public transport fares and toll charges using

information from online sources. All data were organized into tables for clarity, with the workload evenly distributed among members.

AI tools such as ChatGPT and Gemini were used to support our learning and problem-solving process. However, they were used responsibly and critically rather than relied on blindly. For example, calculating carbon emissions for certain routes could be confusing, so these tools helped us understand the formulas and calculation methods involved. In addition, when certain algorithm concepts were difficult to understand, they were used to explain them in simpler terms. Nevertheless, we always verified the information provided and ensured that our final work reflected our own understanding and analysis.

If I were to repeat this project, one improvement I would make is to gather all team members to discuss the problem definition before assigning any tasks. In this assignment, tasks were assigned early, but many members were unable to proceed effectively without clear objectives. In future assignments, I would ensure brainstorming and problem definition are completed first so the objectives are clear from the beginning. I would also encourage more open discussion and communication among team members so everyone feels comfortable sharing their questions and concerns. Through active discussion, members can exchange ideas, clarify doubts, and improve the overall quality and efficiency of the project.

(497 words)

Ayu Wen Li (22017867)

My main contributions to this assignment included collecting data for the carbon emission section, defining the initial state and transition model in the problem analysis section as well as implementing the Uniform Cost-search algorithm. This also included the UCS diagram, process outline and evaluating the suitability of UCS through the evaluation metrics such as completeness, optimality, time complexity, space complexity and practical feasibility.

After completing this assignment, I not only gained a stronger theoretical foundation of AI search algorithms, I was also better able to understand the logic flow behind each path search algorithm and how each stage of the search process conducts evaluations and decisions. By having these stronger graphs of the concepts and process flow, I can better visualise the process behind the algorithms and how it is applied in the AI algorithm's codes. Moreover, after completing the implementation of the UCS sections, I learned more about different evaluation factors which were aspects I had weaker comprehension in before.

In terms of challenges faced during this assignment, I had difficulty in determining how the path cost should be evaluated as there were numerous factors to consider that could potentially influence the cost to the final route such as the carbon emissions, travel expenses, time taken to travel and modes of transportation. As I was tasked with collecting the carbon emission data, I had initially struggled with deciding how carbon emissions should be calculated and the difference in emission factors of various transportation modes such as the LRT, MRT, Urban Bus and Electric Bus (Sunway BRT).

Besides that, I was also unsure about the public transport routes between each state location and how to estimate the distances for each public transportation route which involved multiple stops and transits. In order to solve these problems, I consulted AI tools, specifically Google AI to help estimate the route distances and to find any supporting information behind the outputs given. In order to ensure ethical use and reliability of AI tools, the information was not accepted blindly but was cross-checked against Google Maps for reasonability and consistency.

In terms of group work and communications, we first met physically to discuss the project requirements and clarify our doubts such as how neighbouring nodes should be defined in our assignment context. We also discussed the cost factors to be considered. Questions that were unresolved would later be raised by our group leader to the lecturer of clarification. Subsequent communications were later mostly done through group chat discussions and updates.

Something that I would improve on given the chance would be to standardise the routes early on before data collection begins as this would help in establishing a fixed routes to improve consistency across all calculations and also help to lessen the risk of any potential misalignment in cost, time and carbon emissions especially for the public transportation routes. Overall, this assignment was very helpful in strengthening my foundation of AI search algorithms and how each algorithm may be applied in a real-world scenario.

(500 words)

Keertana A/P Subramaniam (23109614)

As a team member in this group, I contributed to this project in several key components of the system design and analysis. I took responsibility to define and understand the goal state, developing the path cost function along with my group leader, implementing the Greedy Best-First Search algorithm although it wasn't included in this project, and writing the conclusion and justification section.

This assignment has helped me develop a very clear understanding of AI search algorithms and how they can be applied to real-world optimization problems. By defining the goal state, I learnt the importance of multi-goal optimization since the objective of this project was to ensure that all the locations had to be visited at least once where the root node starts from Sunway University. This section made me think and understand how AI search problems can involve multiple termination conditions. This section required me to ensure the rest of the problem more systematically.

As for the path cost function, I learnt how to combine different factors of the real-world into a single evaluation metric. By balancing both carbon emissions and travel expenses, normalization was required so that both the values were contributed to the final cost fairly. This section was very important as it demonstrated how AI models rely heavily on feature scaling and weighting that is correct so that biased decision-making can be avoided.

Although BFS, UCS and A*Search algorithms were chosen for this project to be evaluated, I was assigned to explore the Greedy Best-First Search so that it can be compared with the rest of the algorithms and chose the best 3 from it. This algorithm helped me understand how decision-making based on heuristic-only behaves compared to algorithms like A* that leans more towards cost awareness as well. By comparing both this and A*Search algorithm, it further strengthened my understanding of why A*Search is much more reliable for optimization problems.

I was able to comprehend the overall objective of the project while doing the conclusion and justification section. When comparing all the algorithms chosen side by side, it helped me see that the project did not only find a best route but also justified which algorithm performed the best based on the problem constraints, efficiency, optimality, and real-world applicability. This helped me realise that even small differences in distance, carbon emission, and travel expenses calculations can affect the final comparisons because it is important for the dataset, assumptions, and evaluation criteria to be consistent.

One thing I would improve for this assignment is that I would focus on aligning all the assumptions and intermediate calculations early on in the project. I believe this can help reduce the inconsistencies. Thus the comparison made between the algorithms will be smoother and more efficient.

In general, this assignment improved my comprehension of AI search algorithms and how they can be applied in a real-world situation while considering sustainability as well. I also developed better analytical thinking while I was evaluating the differences in algorithms.

(496 words)

Siow Qi Yung (22053037)

In this assignment, I contributed to creating the state space by constructing a graph, where each node represents a residence and each edge represents the cost between residences, considering both carbon emission and travel expenses. I also helped in creating the transition model, which shows how the search moves from one residence to another. I also contributed to the Breadth-First Search (BFS) algorithm where I built a search tree diagram for BFS to visualize the search process and identify possible solution paths. I also evaluated the algorithm based on completeness, optimality, time complexity, space complexity, practical feasibility and sustainability considerations.

Through this assignment, I learned that AI search algorithms can be used to solve route-planning problems by systematically exploring possible paths. This helped me realize that selecting an appropriate algorithm is important because different problems may require different searching methods. I gained a better understanding of concepts such as state spaces, transition models, search trees and goal states. I also learned that different search algorithms have different advantages and limitations. For example, BFS guarantees finding the best path with fewest steps, but it may not always find the route with the lowest cost. This assignment also helped me understand how AI search techniques are applied in the real-world such as finding the shortest route on a map.

One of the challenges we faced was understanding how to represent the problem as a graph and deciding how the residences should be connected. Initially, our group was unsure how the path costs should be assigned. I also found it difficult to understand how BFS explores nodes level by level. Drawing the search tree diagram and identifying loop nodes were also challenging because I needed to ensure that the search process was expanded correctly. However, through discussion with my group members, we finally came out with solutions to address the challenges we faced. I also kept reviewing lecture notes and other online materials to understand BFS concepts and other related knowledge. These challenges helped to improve my problem-solving skills.

Our group collaborated through physical meetings and online discussions. The group leader divided tasks among members, shared ideas and reviewed each other members' work to ensure our work meet the assignment requirements. When we encountered difficulties, we discussed together and referred to lecture materials to clarify our understanding.

I have used AI tools such as ChatGPT responsibly during this assignment. I used the tool mainly to improve my understanding of transition model, BFS and other AI concepts. I did not copy the generated content directly. Instead, I used it as a learning resource and rewrote the explanations in my own words.

If I had the opportunity to improve this project, I would include more factors when calculating the path cost, such as travel time to make the route planning more realistic. I would also like to explore more complex route-planning scenarios to gain a deeper understanding of AI search techniques.

(486 words)

Vidhipriya Devi Pokiri (23057953)

As a member of the Gigggle group, my contributions included writing the introduction section, collecting distance data, and defining the travel actions. I was also responsible for implementing the Depth-First Search (DFS) algorithm and carrying out the final algorithm comparison section at the end of the project.

Writing the introduction helped me set the tone by highlighting why AI route planning is so essential for both sustainability and efficiency. With that foundation in mind, I mapped out the routes between our residential areas using Google Maps for the distance data section. I also categorized our movement options in the actions section, breaking them down into walking, public transport, and driving. This part of the project was highly valuable as it taught me how to take everyday physical logistics and translate them into a structured mathematical format that an AI model can easily process.

One of the more interesting challenges I faced was shifting my focus to the Depth-First Search (DFS) algorithm. Coding this was a unique task because the algorithm is designed to push forward down a single path until it reaches the goal or hits a dead end, only backtracking to explore other branches when it has no other choice. Because DFS is an uninformed search strategy, I had to strip away all the data regarding path costs. This meant the algorithm completely ignored our weighted variables for carbon emissions and travel expenses. Starting from Sunway University, it simply picked neighboring nodes arbitrarily, indifferent to the cumulative costs of the trip. Seeing this in action was a great learning experience. It visually demonstrated how DFS prioritizes path depth over actual efficiency, proving exactly why it was not the right fit for our specific optimization needs.

Evaluating the practical applications of different search strategies in the algorithm comparison section brought the whole assignment together for me. By analyzing BFS, UCS, and A* Search side by side, I gained a much deeper understanding of how algorithms behave differently under specific constraints. I was able to determine that A* Search was the most effective method for balancing our travel expenses and environmental factors. This analysis showed me that the project wasn't just about finding a route, but critically justifying which method performs best for real-world applications.

If I were to repeat this project, one improvement I would make is to spend more time exploring how different heuristic values might further optimize our search results from the beginning. In general, this assignment significantly improved my understanding of AI search algorithms and

how they can be used to solve everyday problems while keeping sustainability in mind. I also developed stronger analytical skills by actively evaluating the structural limitations and strengths of different problem-solving methods.

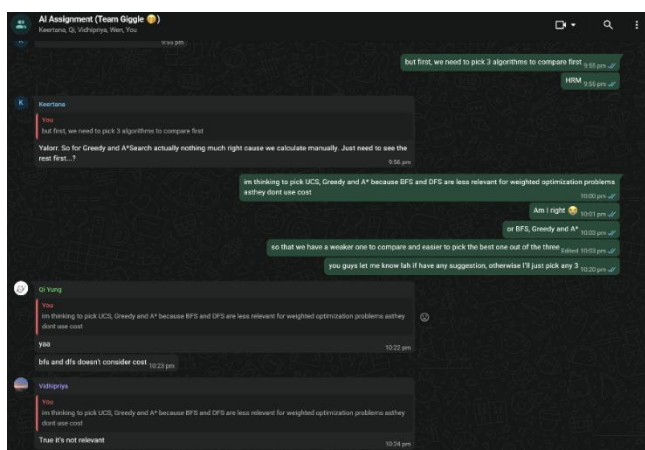
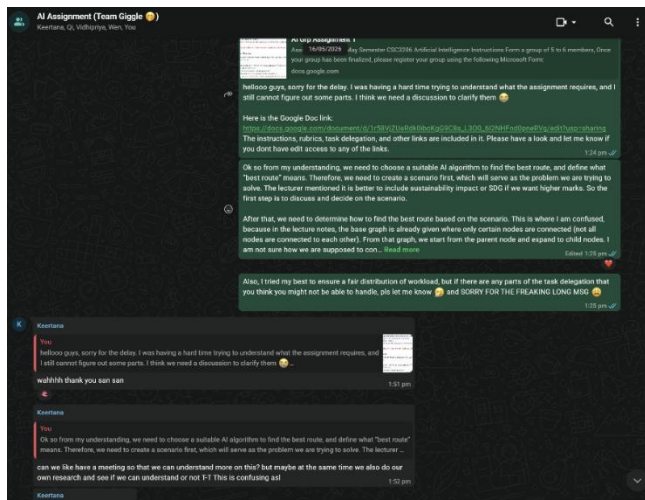
(466 words)

5.3 Evidence of Group Collaboration

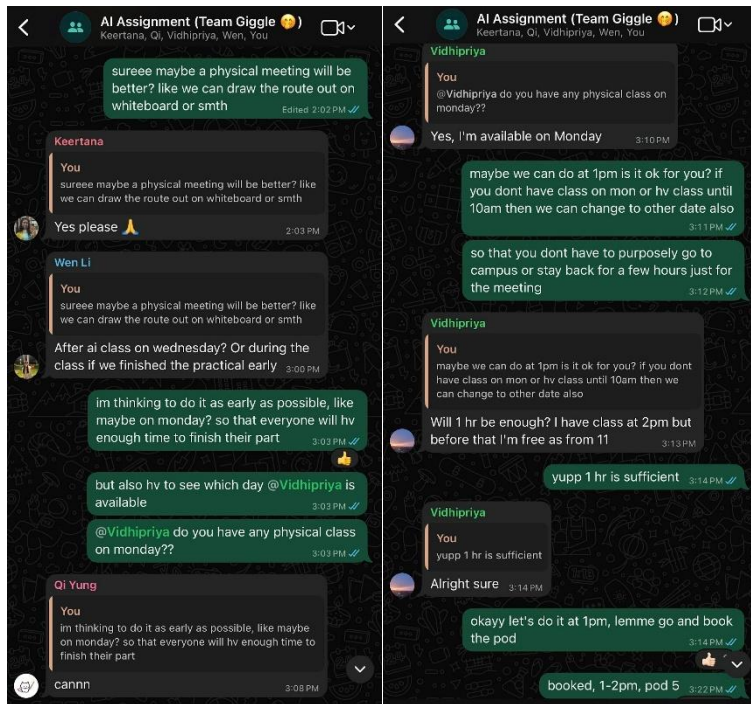
Physical Meeting:



Discussions:



Scheduled for physical meeting:



Planning documents - Task allocation records:

Task	PIC	Internal Due	Status
1.0 Introduction	Vidhipriya	19/5	☑
2.0 Problem Definition and Data Collection			☑
2.1 Problem Definition/Scenario	Hui San	19/5	☑
2.2 Data Collection			☑
2.2.1 Distance Data	Vidhipriya	21/5	☑
2.2.2 Travel Time Data	Keertana	21/5	☑
2.2.3 Carbon Emission	Wen li	21/5	☑
2.2.4 Cost	Hui San	21/5	☑
3.0 Problem Analysis (AI Formulation)			☑
1) State space	Qi Yung	23/5	☑
2) Initial state	Wen li	23/5	☑
3) Goal state	Keertana	23/5	☑
4) Actions	Vidhipriya	24/5	☑
5) Transition Model	Wen li	24/5	☑
6) Path Cost Function	Hui San	24/5	☑
	Keertana		
4.0 AI Algorithm Evaluation			☐
Try out all algorithms, and choose the top 3 algorithms that generate the best path	ALL	28/5	☑
a) BFS	Qi Yung	28/5	☑
b) DFS	Vidhipriya	28/5	☑
c) Uniform Cost	Wen li	28/5	☑
d) Greedy	Keertana	28/5	☑
e) A search	Hui San	28/5	☑

4.1 BFS	Qi Yung	30/5	☑
4.2 UCS	Wen li	30/5	☑
4.3 A* Search	Hui San	30/5	☑
5.0 Algorithm Comparison and Conclusion			☑
5.1 Algorithm Comparison	Vidhipriya	2/6	☑
5.2 Conclusion and Justification	Keertana	2/6	☑
7.0 Evidence of Group Collaboration (Appendix)	ALL	3/6	☐
8.0 AI Usage Transparency (Appendix)	ALL	3/6	☐
9.0 Individual Reflection (Appendix)	ALL	5/6	☐
Transfer everything to MS word doc	ALL	8/6	☐
Check page limit and similarity rate	Hui San	8/6	☐
6.0 Presentation Video			☐
Problem Definition and Data Collection	Keertana	9/6	☐
Problem Analysis	Vidhipriya	9/6	☐
BFS Evaluation	Qi Yung	9/6	☐
UCS Evaluation	Wen li	9/6	☐
A* Search Evaluation	Hui San	9/6	☐
Conclusion with justification	Keertana	9/6	☐
Video Editing (Just compile all recordings)	Qi Yung	10/6	☐
Final check	ALL	11/6	☐
Submit on elearn	Hui San	12/6	☐

5.4 AI Usage Transparency

Bus Route Fair Query:

<https://chatgpt.com/share/6a104c57-7d40-83ec-97dc-48b10f1d5480>

Carbon Emission Query:

https://www.google.com/search?smstk=ChhNbWs3Qyt0WlkxZkx6ZmRWY0FpZEsxST0QAQ%3D%3D&smstidx=0&q=how+is+average+car+emission+factor+calculated+in+malaysia&udm=50&csuir=1&aep=34&kgs=7c8f35a3b8b1e8f9&shem=rims pwouoe&shndl=37&shmd=H4sIAAAAAAAAA_3WOO24CMRRFM11EE4kylUWJIDE0SCD2Y11sYz_Jn9F7hoGOZbEGVkSZiZRUEe2Rzj139uxm21hHRaJw9ozglQUrn0mEalFH2FZ5YsmeEpp3iorKSLgK4XMfWxtkp_U4jn2Qhka2tzVr8WAbvwauWpQNhLB3rR4yocCSv1QwmL-Z4daQ_I9SC_fLrePW3d_f3Tm1Tb16aXoA6M4KuHXDiYwHPnS9Pryj5nJLw7szHqzcsNP_hvHeKj3-wAAAA&shmds=v1_ATWGeeOxQT41gjdXa5nOkHnyY-mahmY2_a48wt42gmFhzvOeWg&source=sh%2Fx%2Faim%2Fm5%2F1

Transition Model:

<https://chatgpt.com/share/6a154bf7-ebac-83ec-874f-eb5fd508bfa7>

DFS Query:

<https://chatgpt.com/share/6a3124d0-27c8-83ec-8f5e-8f64a671f96e>

6.0 References

- Asian Transport Observatory. (2026). *Malaysia Transport and Climate Profile 2026*. Asian Transport Observatory. <https://asiantransportobservatory.org/analytical-outputs/transportclimateprofiles/malaysia-transport-and-climate-profile-2026/>
- GeeksforGeeks. (2025, July 22). *A* algorithm and its Heuristic Search Strategy in Artificial Intelligence*. GeeksforGeeks. <https://www.geeksforgeeks.org/artificial-intelligence/a-algorithm-and-its-heuristic-search-strategy-in-artificial-intelligence/>
- GeeksforGeeks. (2024, August 23). *Uniform Cost Search (UCS) in AI*. GeeksforGeeks. <https://www.geeksforgeeks.org/artificial-intelligence/uniform-cost-search-ucs-in-ai/>
- Jax. (2025, September 25). *What is a good fuel consumption per 100km*. PCauto. <https://www.pcauto.com.my/faq/what-is-a-good-fuel-consumption-per-100km-faq1105>
- Kumari, S. (2025). *Informed Search Algorithms in Artificial Intelligence*. Code360 by Coding Ninjas. <https://www.naukri.com/code360/library/informed-search-algorithms-in-artificial-intelligence>
- Prasarana Malaysia Berhad. (n.d.-a). *Rapid KL - BRT*. MyRapid. <https://myrapid.com.my/bus-train/rapid-kl/BRT/>
- Prasarana Malaysia Berhad. (n.d.-b). *Rapid KL - Integrated Fare Table*. MyRapid. <https://myrapid.com.my/bus-train/rapid-kl/integrated-fare-table/>
- Prasarana Malaysia Berhad. (n.d.-c). *Web Journey Planner*. MyRapid. <https://myrapid.com.my/journey-planner/>